A closer look into the transformer architecture

April 20, 2026

Did you complete the reading assignments?

Estimating (GPU) RAM usage of a LLM

- I have a 7 billion parameter model. How much RAM do I need to host it?

- RAM Usage = 7 billion params x 2 bytes per parameter = 14 billion bytes*
  - **Kilo** bytes = 2^{10} = 1,024 ≈ **1 thousand bytes**
  - **Mega** bytes = 2^{20} = 1,048,576 ≈ **1 million bytes**
  - **Giga** bytes = 2^{30} = 1,073,741,824 ≈ **1 billion bytes**
  - **Tera** bytes = 2^{40} = 1,099,511,627,776 ≈ **1 trillion bytes**

- Unlike regular ML models where a parameter is 4 bytes (32 bit float), LLMs are usually 2 bytes

* Actually we need a bit more for the intermediate computations (or a lot more for long context)

Recap: How a transformer works

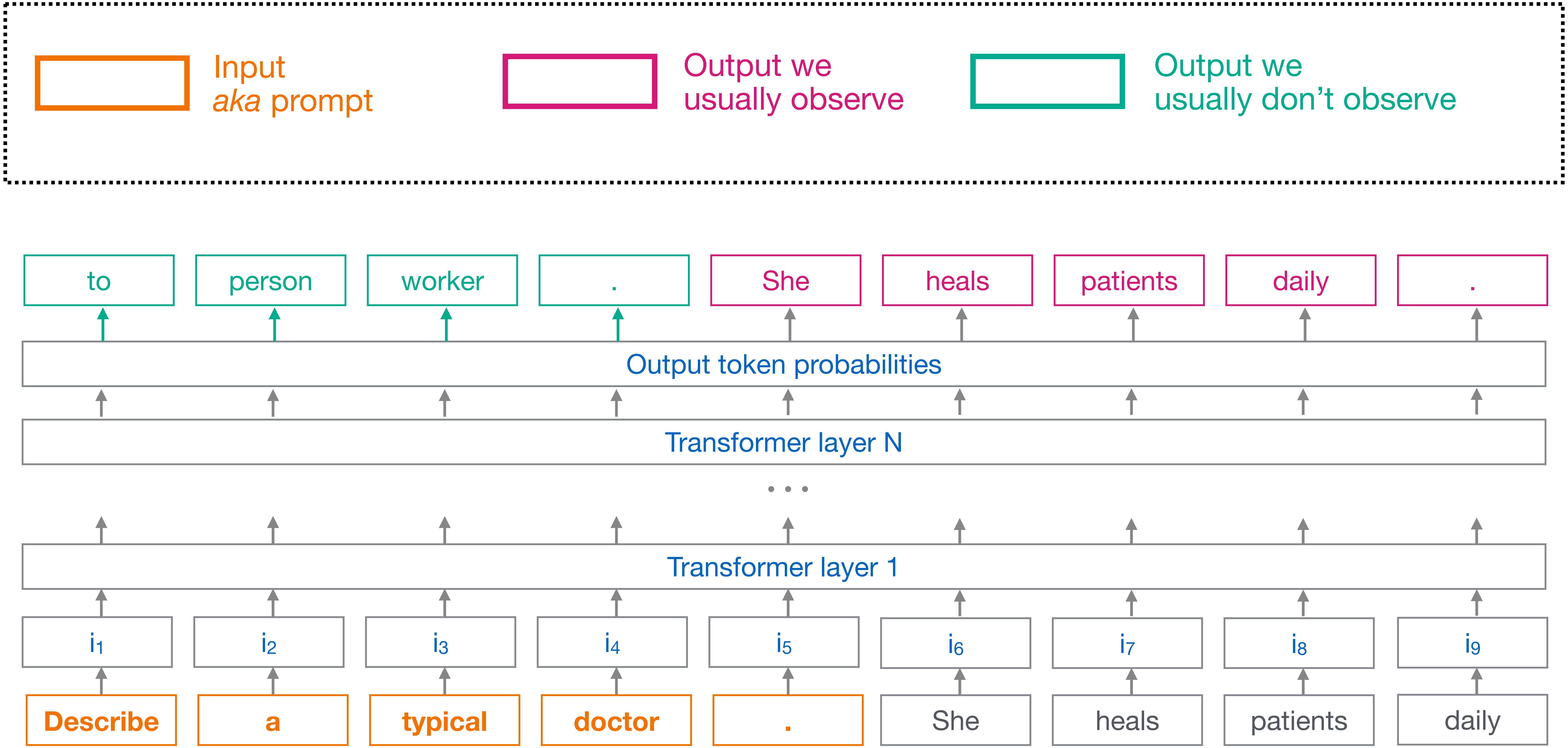
She heals patients daily.

Large Language Model

Describe a typical doctor.

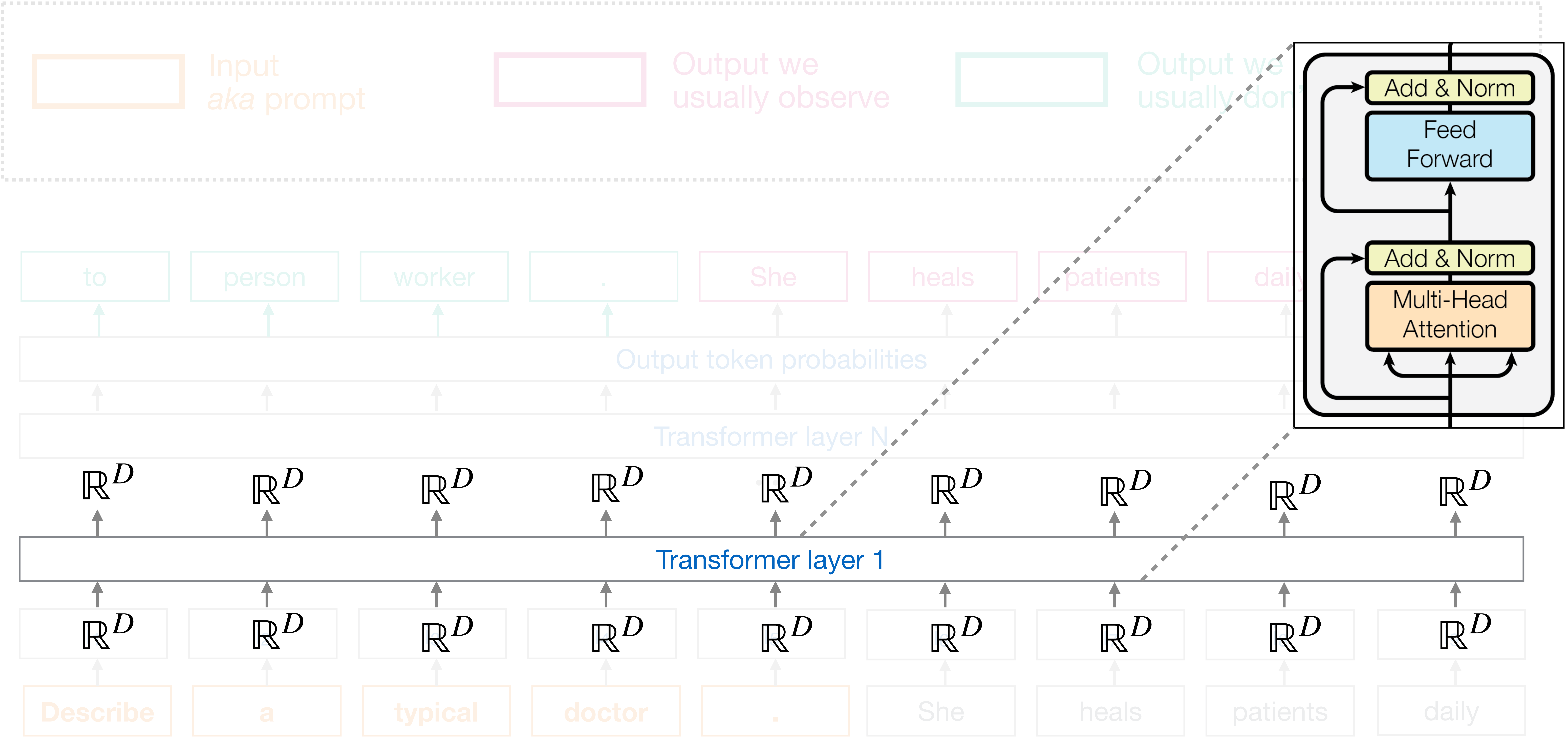
*Fictitious Example

Recap: LLMs and causal language modeling



Exercise

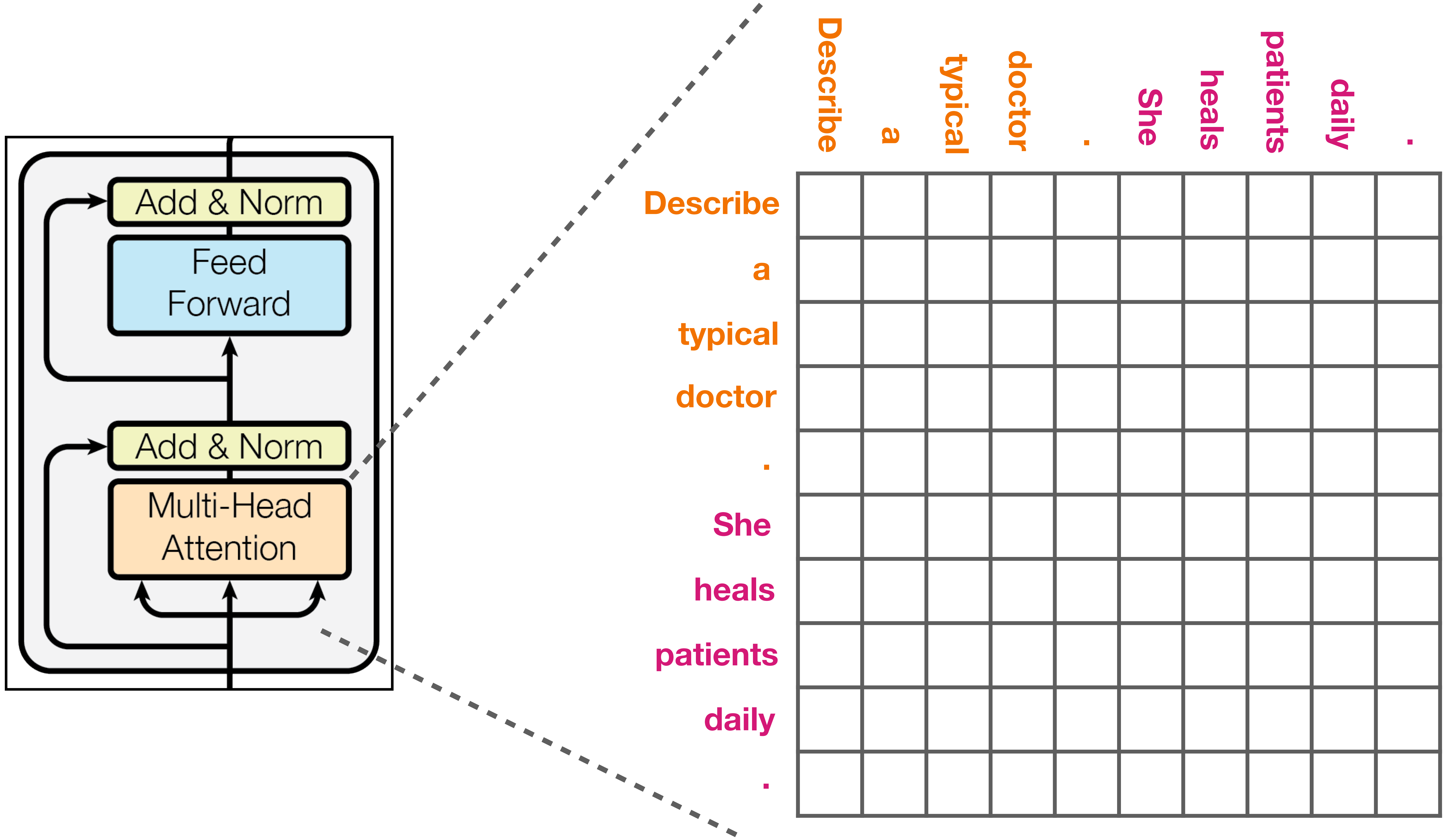
Recap: LLMs and causal language modeling



[Vaswani et al.]

Self attention between all token pairs

When processing *token i*, how much each *token j* should contribute?



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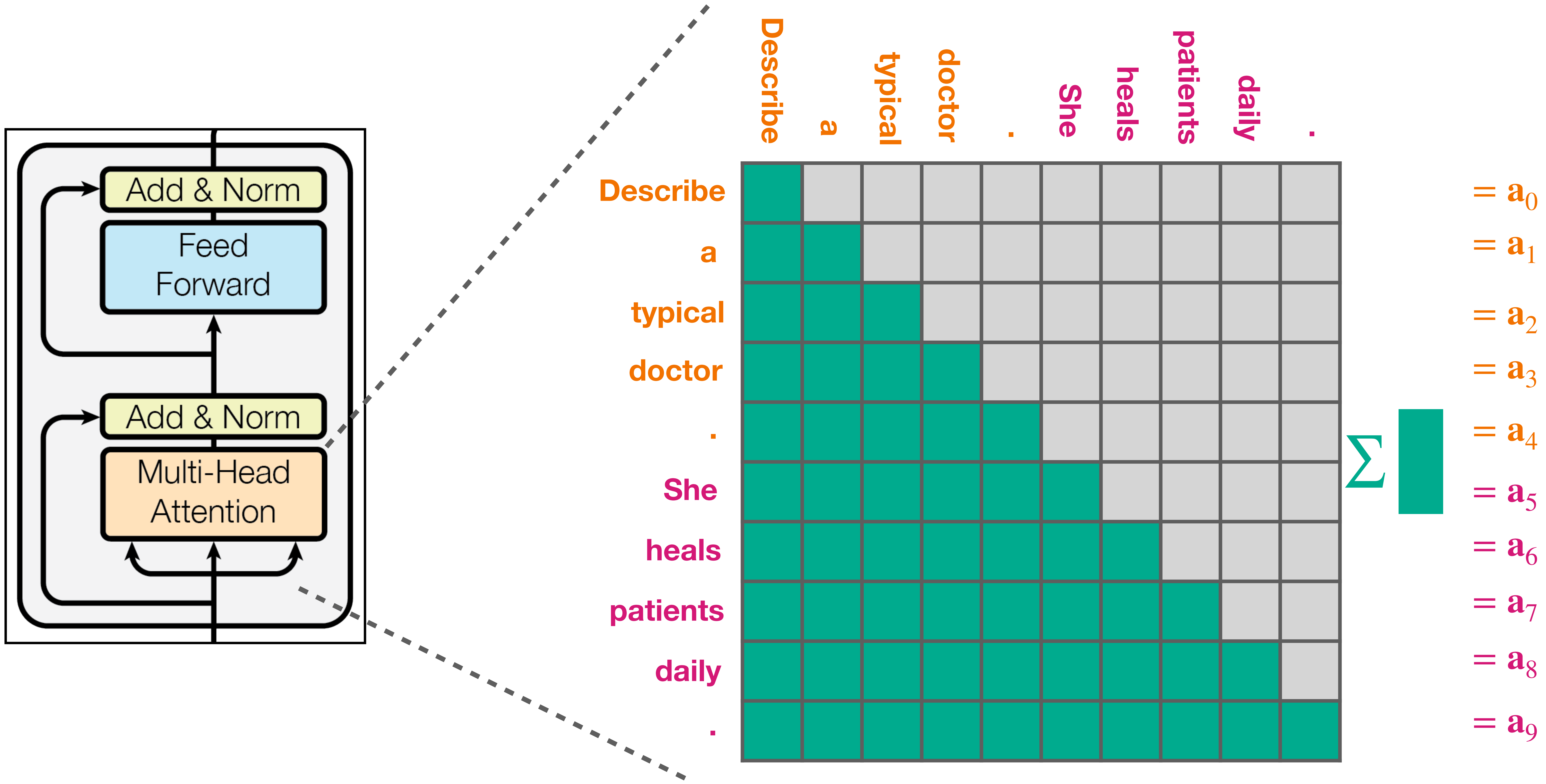

Causal LLMs

Attention of *token i* depends only on *current or previous* tokens

$$\mathbf{a}_i = \sum_{j \leq i} \text{attn}(i, j)$$

[Vaswani et al.]

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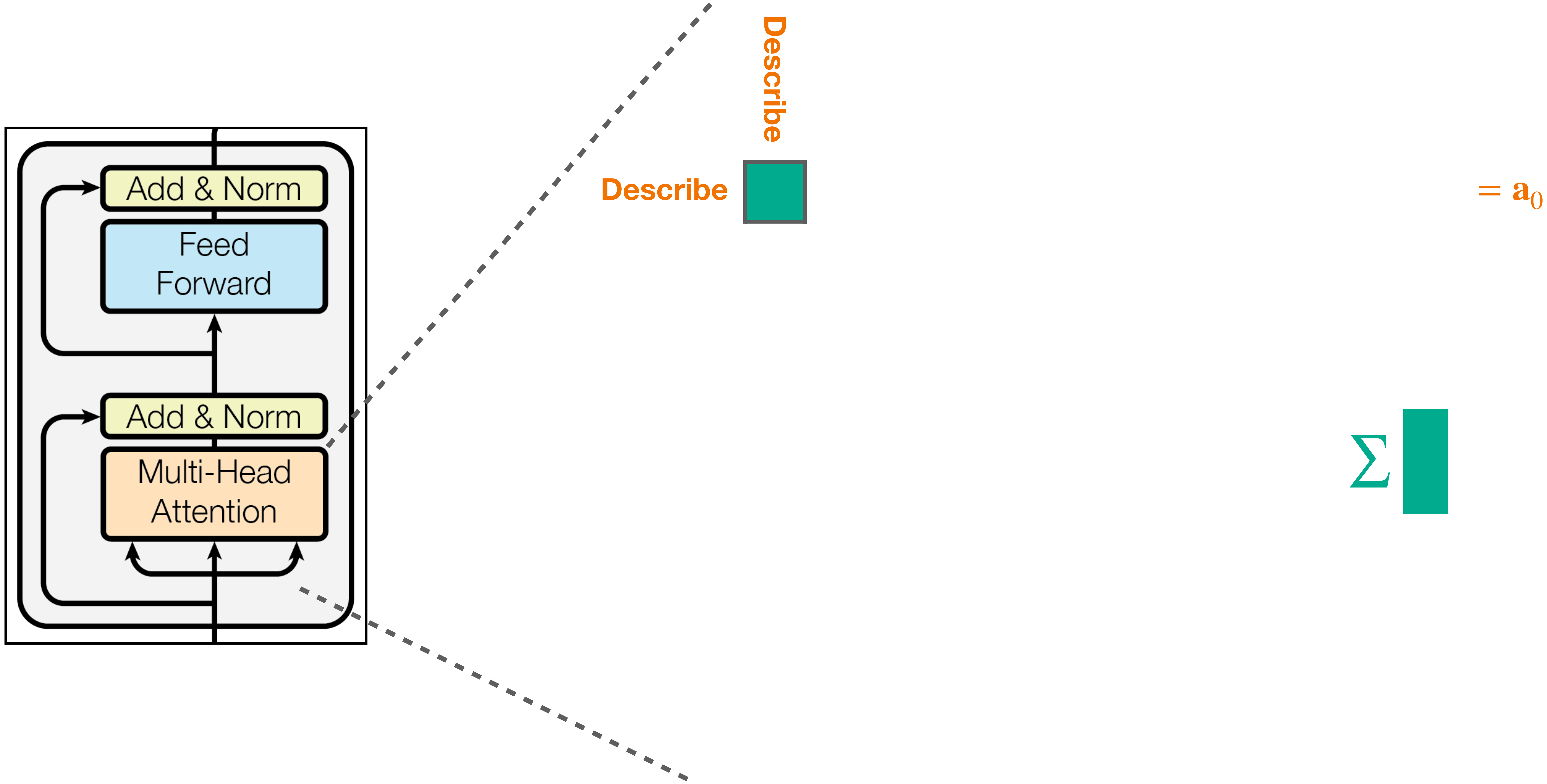

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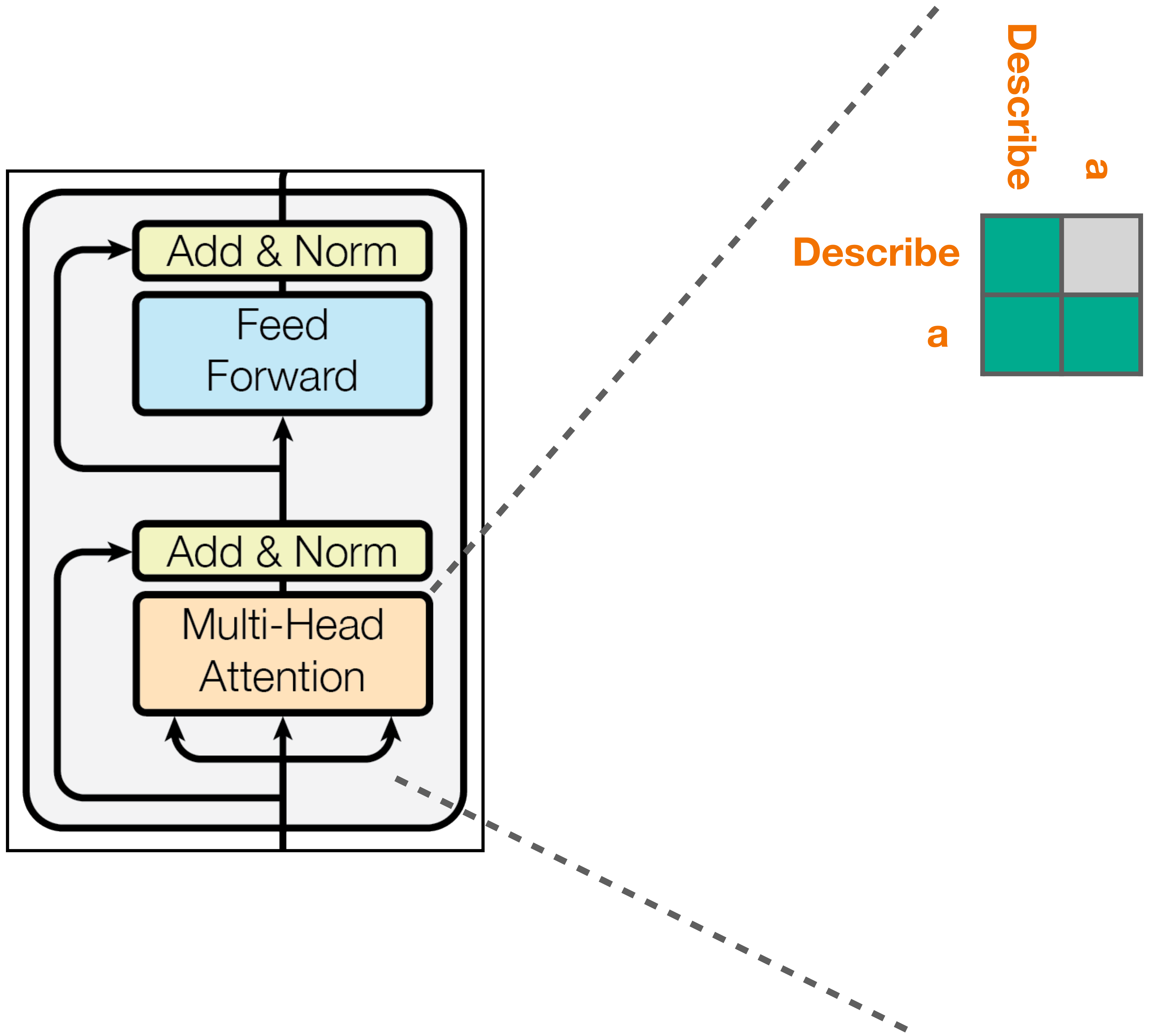

$= \mathbf{a}_0$

$= \mathbf{a}_1$

$\sum$

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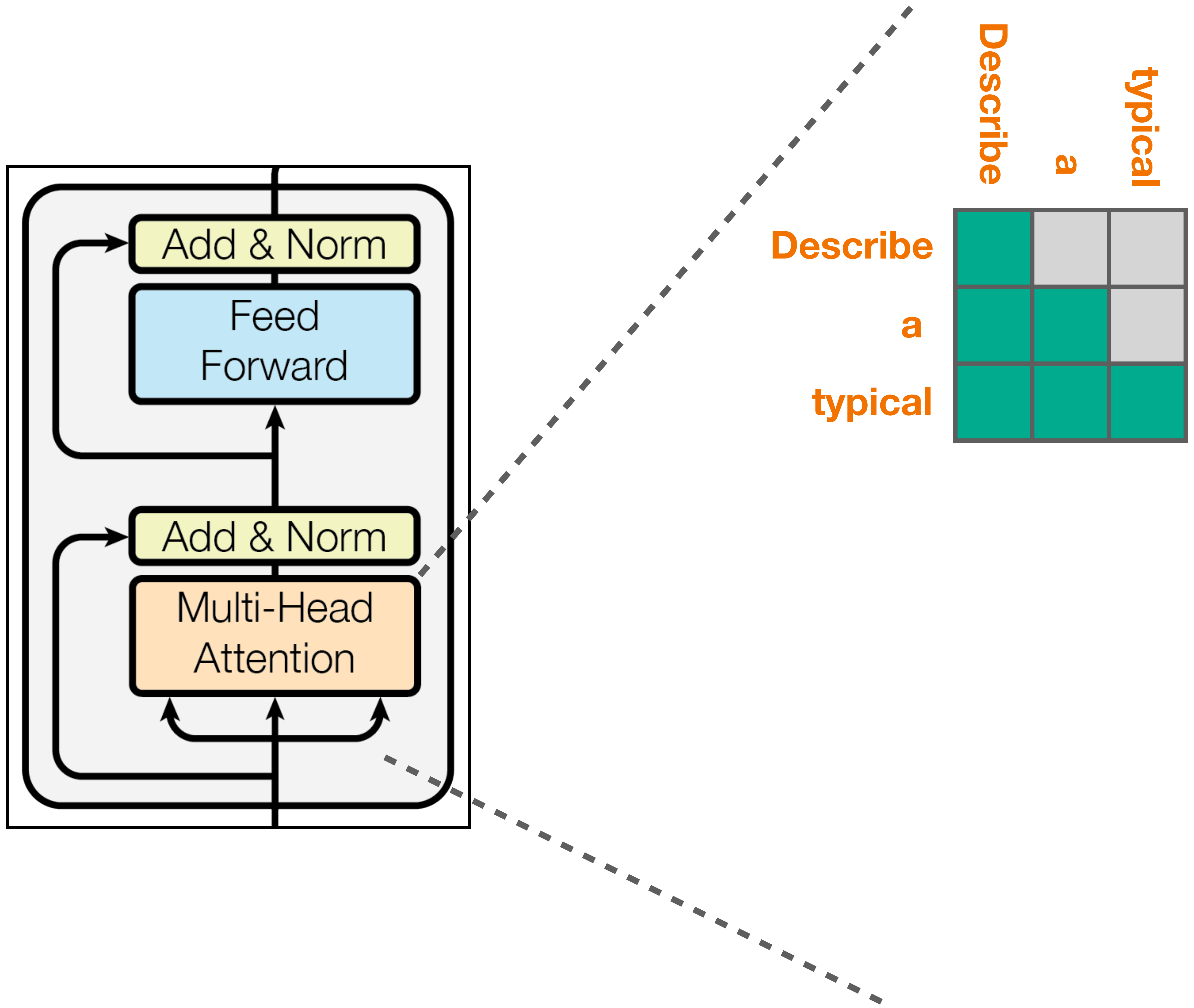

$= \mathbf{a}_0$

$= \mathbf{a}_1$

$= \mathbf{a}_2$

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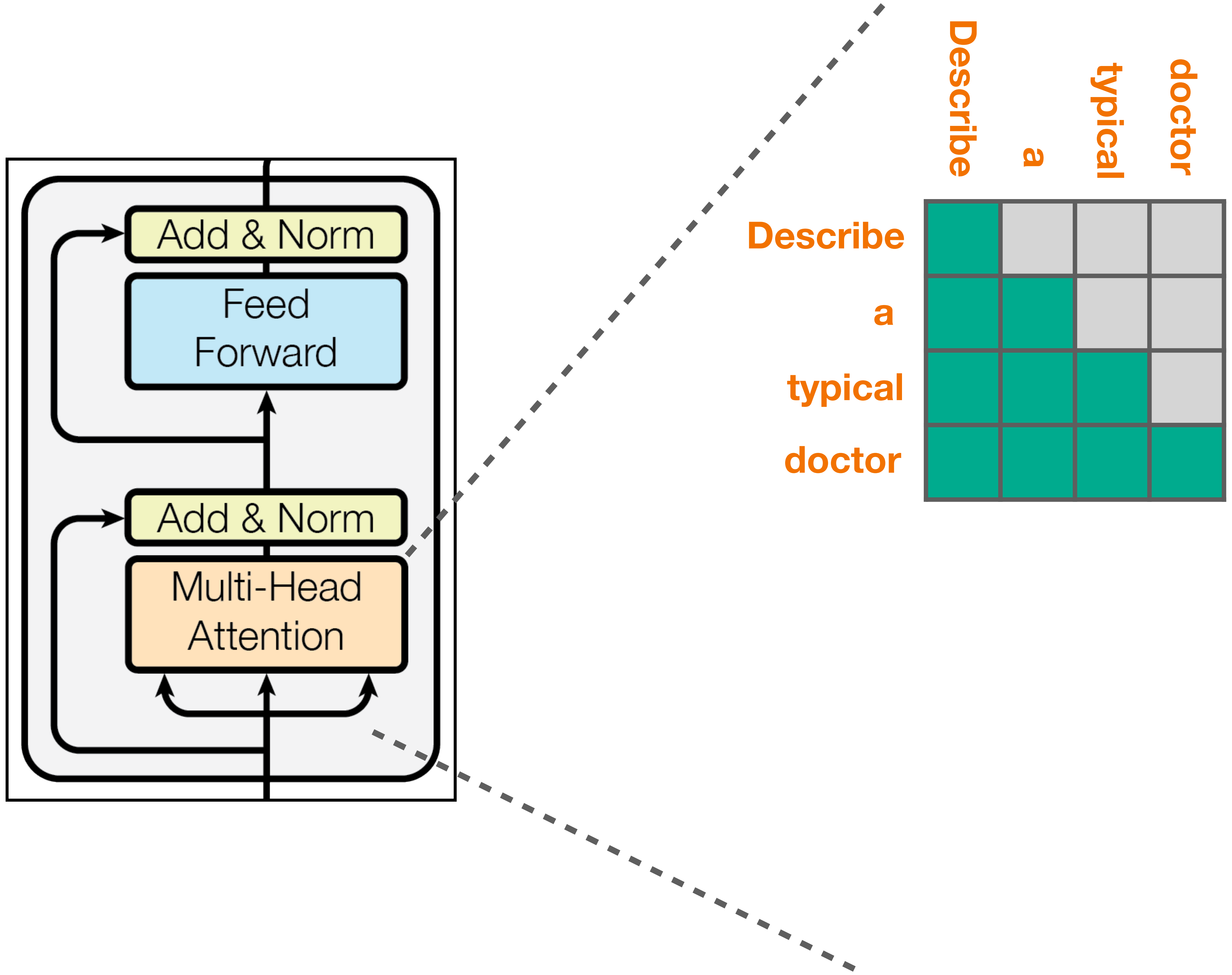

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$= \mathbf{a}_1$

$= \mathbf{a}_2$

$= \mathbf{a}_3$

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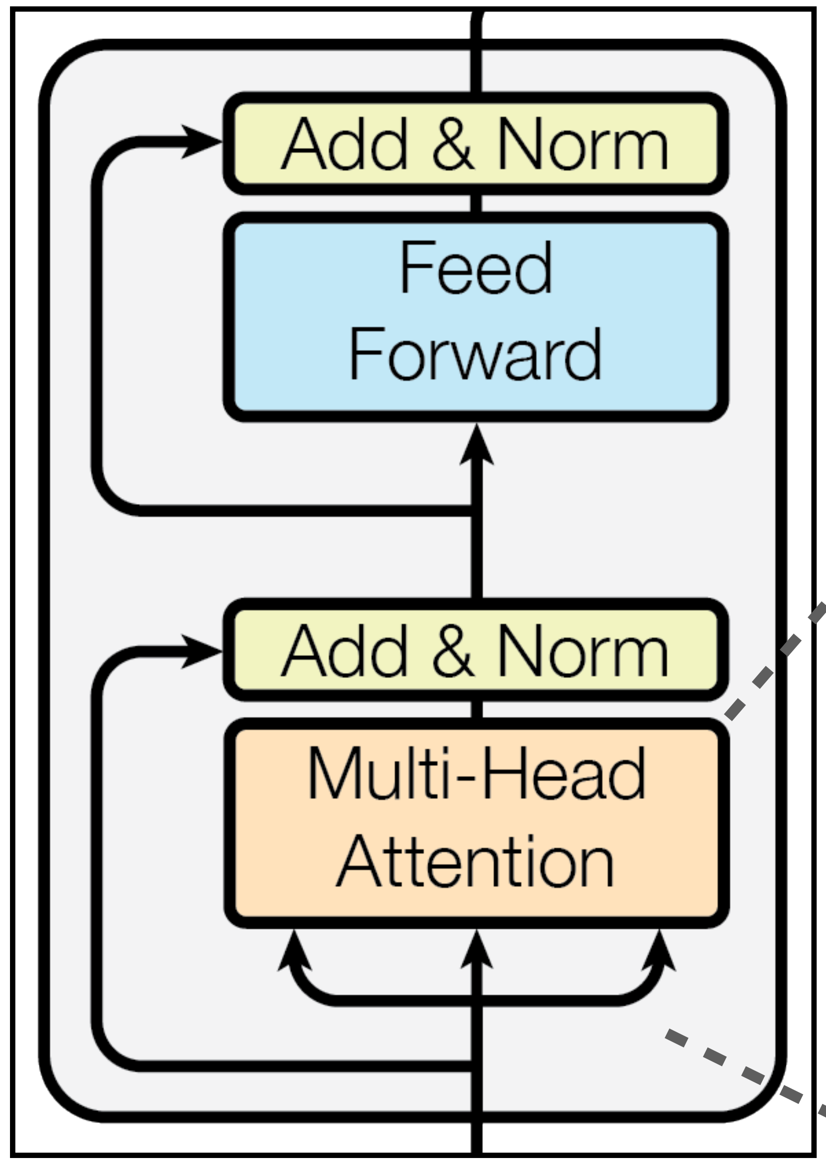

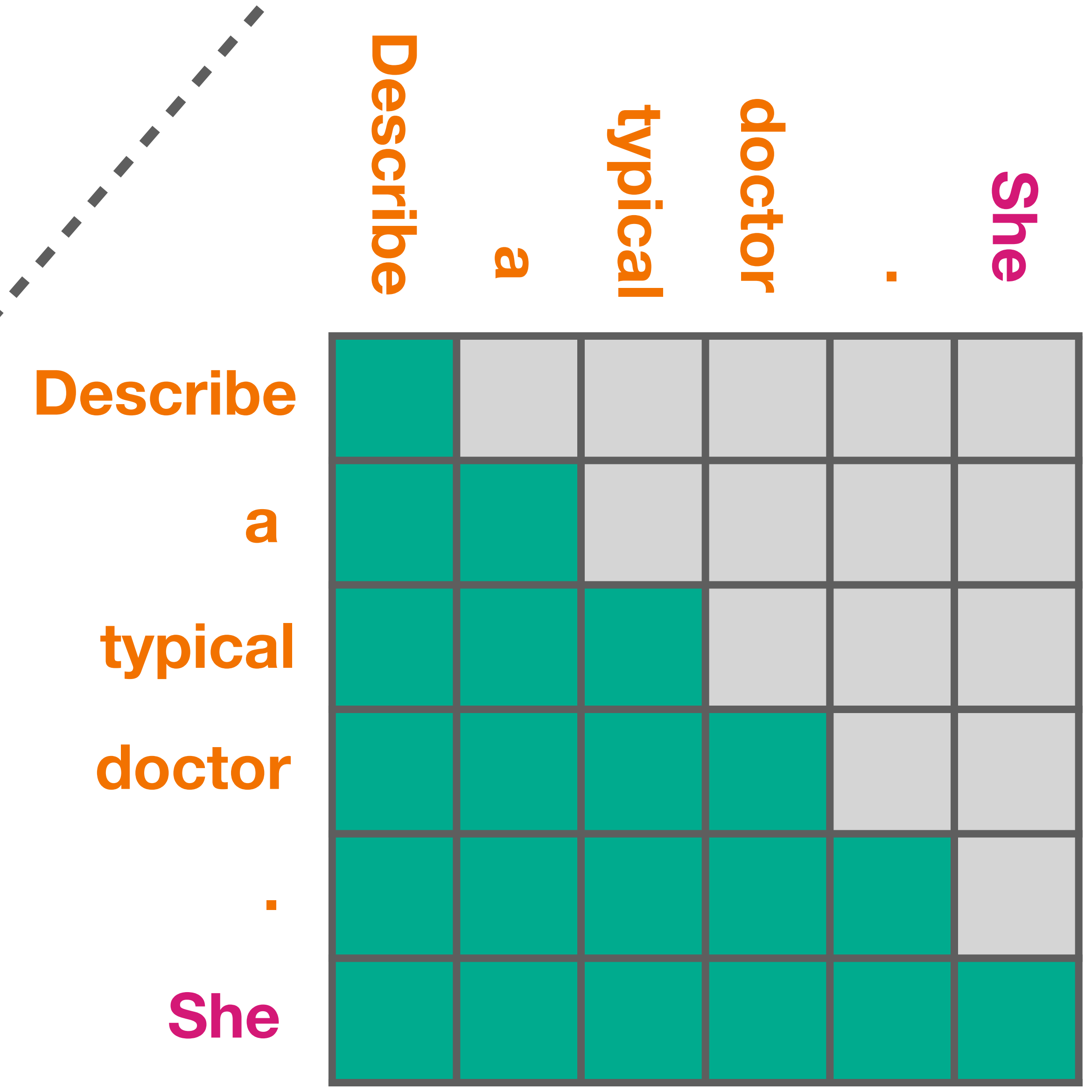

$= \mathbf{a}_0$
$= \mathbf{a}_1$
$= \mathbf{a}_2$
$= \mathbf{a}_3$
$\sum \quad = \mathbf{a}_4$
$= \mathbf{a}_5$

Causal LLMs

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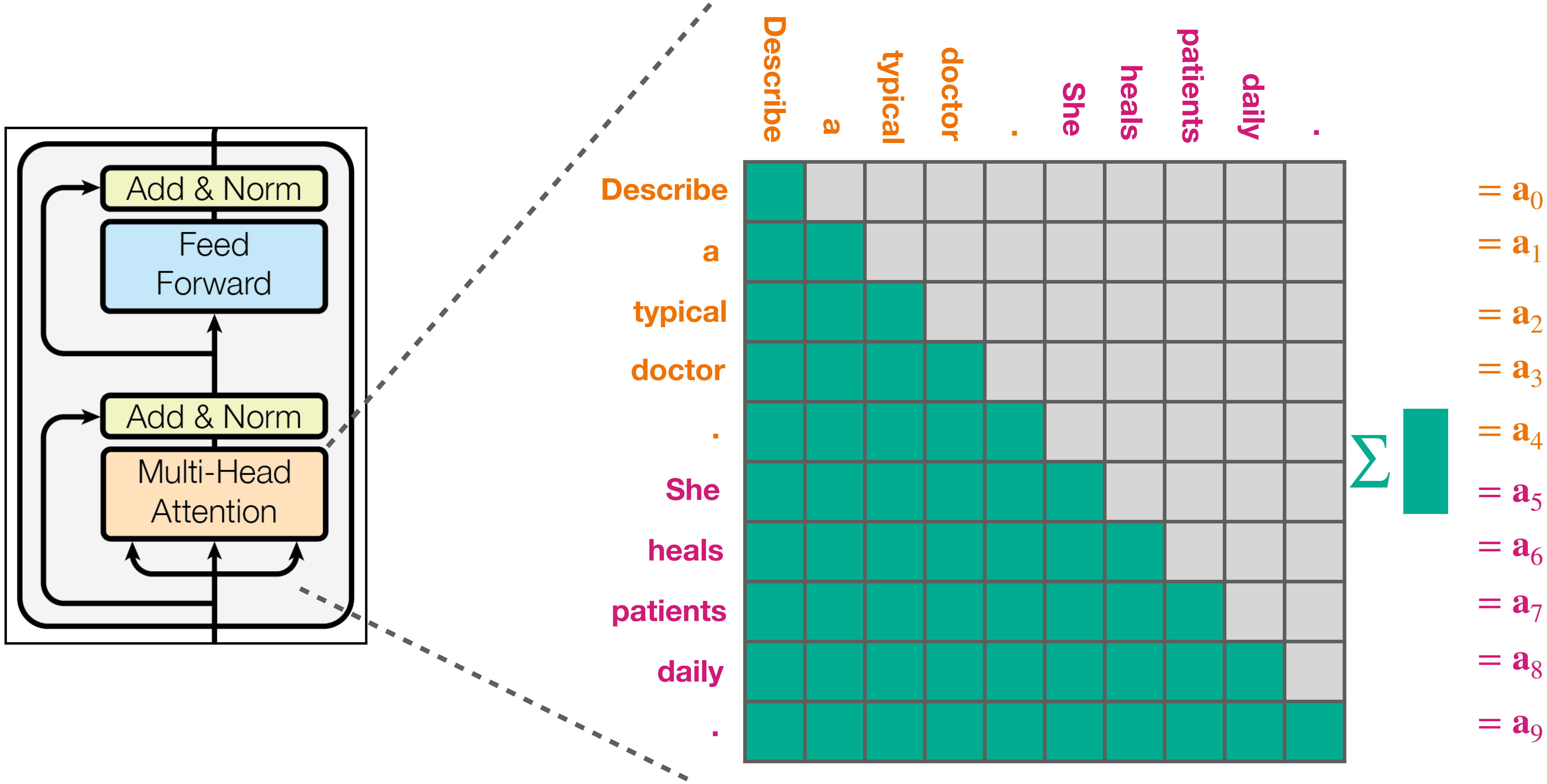

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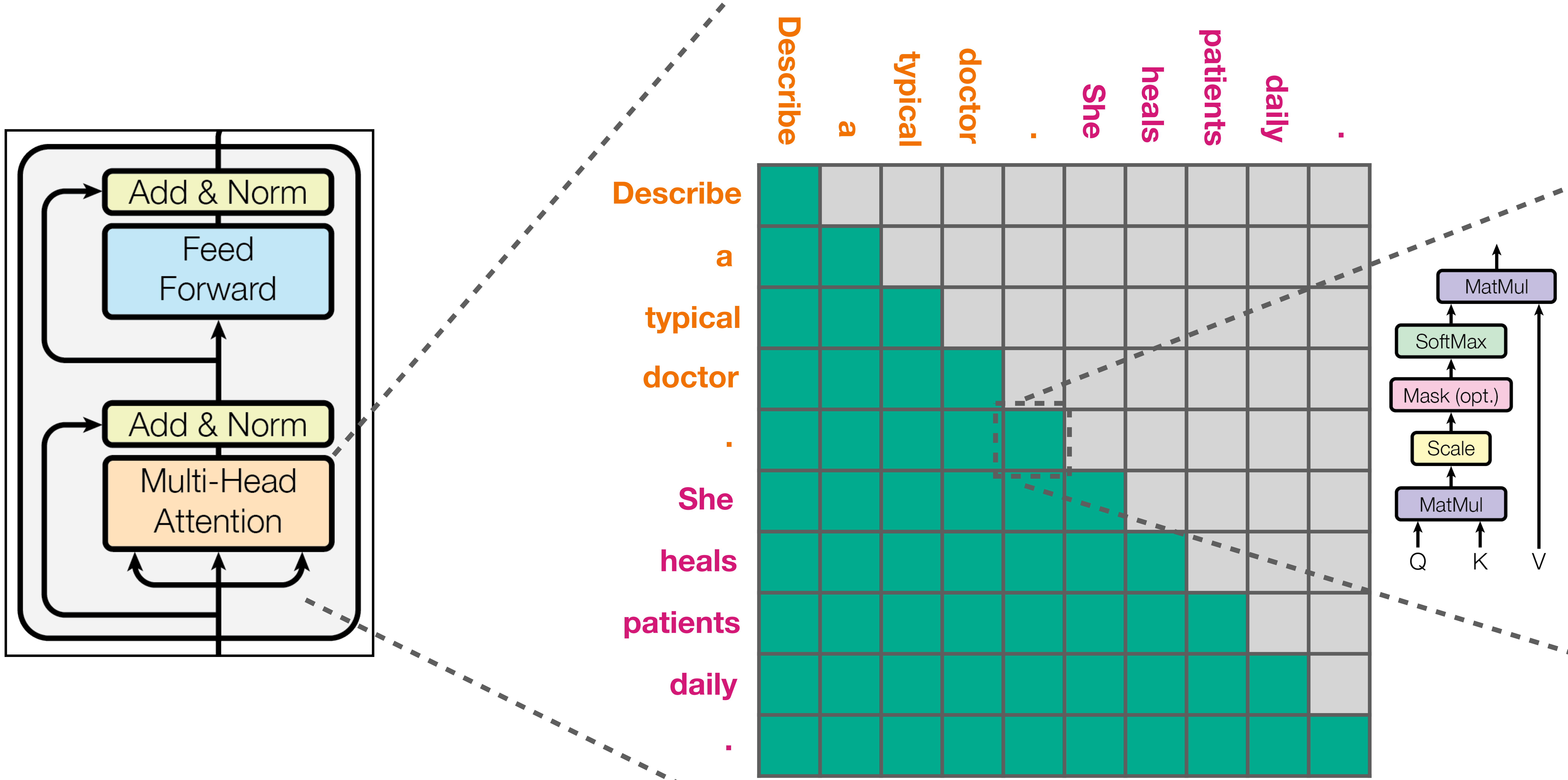


$$\text{attn}(i,j) = \text{softmax}\left(\frac{\mathbf{q}_i \mathbf{k}_j}{\sqrt{d_k}}\right)\mathbf{v}_j$$

$$\text{attn}(i,j) \in \mathbb{R}^D$$

$$\mathbf{a}_i = \sum_{j \leq i} \text{attn}(i,j)$$

[Vaswani et al.]

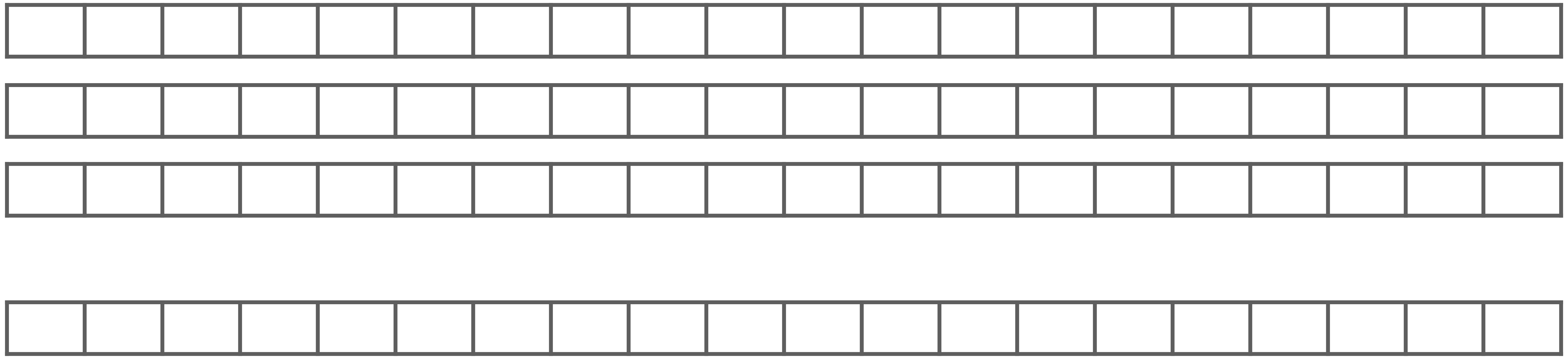
The nitty gritty

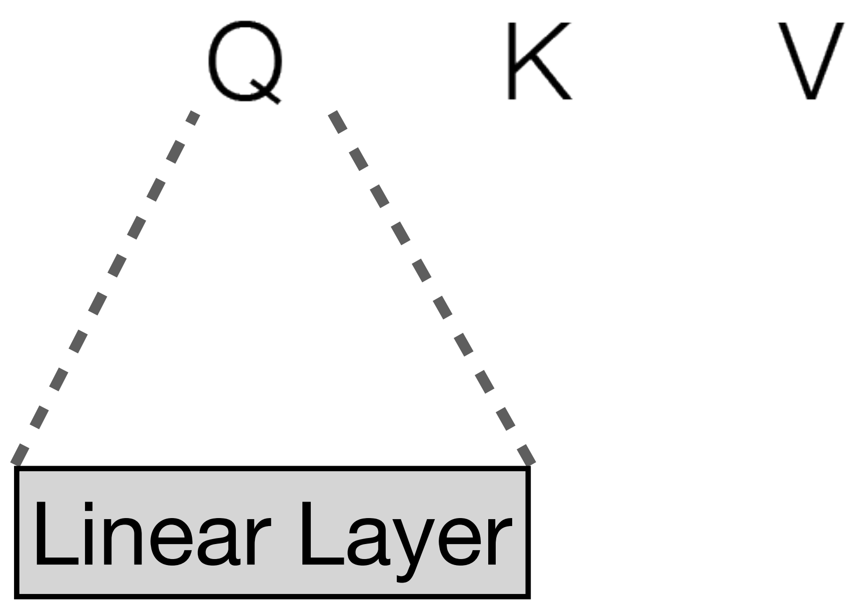
$\mathbf{q}_i \in \mathbb{R}^D$

$\mathbf{k}_i \in \mathbb{R}^D$

$\mathbf{v}_i \in \mathbb{R}^D$

$\mathbf{t}_i \in \mathbb{R}^D$





Vaswani et al.

The nitty gritty

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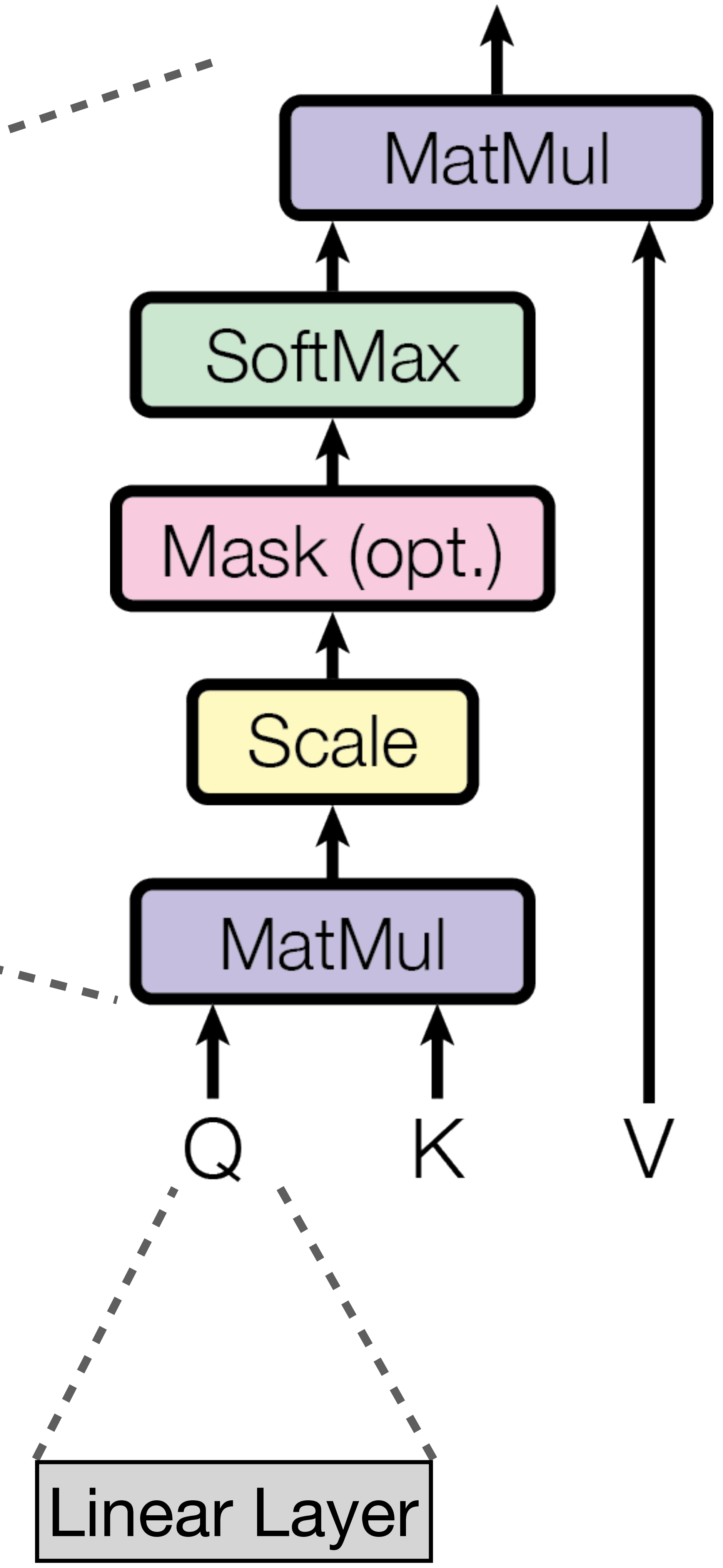


$\mathbf{q}_i \in \mathbb{R}^D$

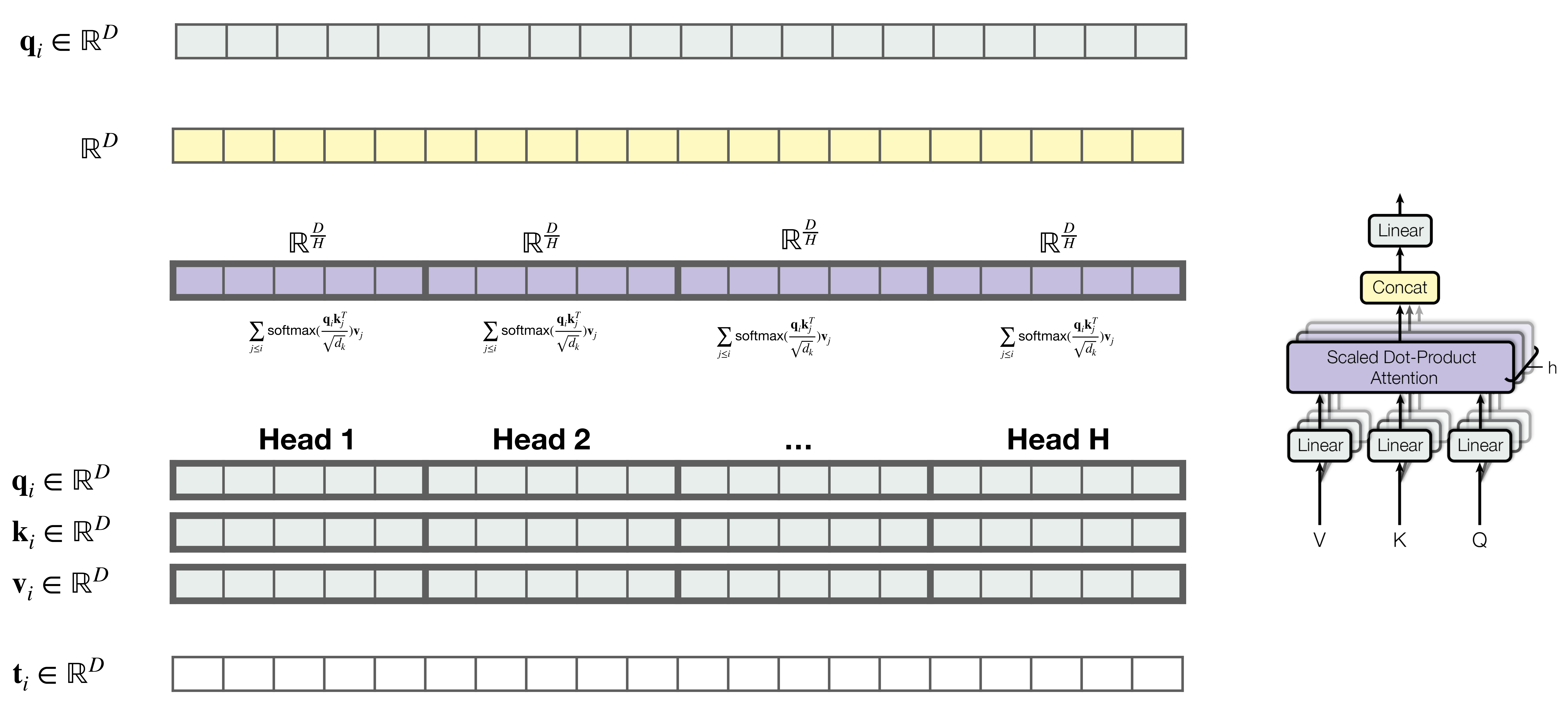
$\mathbf{k}_i \in \mathbb{R}^D$

$\mathbf{v}_i \in \mathbb{R}^D$

$\mathbf{t}_i \in \mathbb{R}^D$

Vaswani et al.

Attention is multi-headed



The full transformer layer

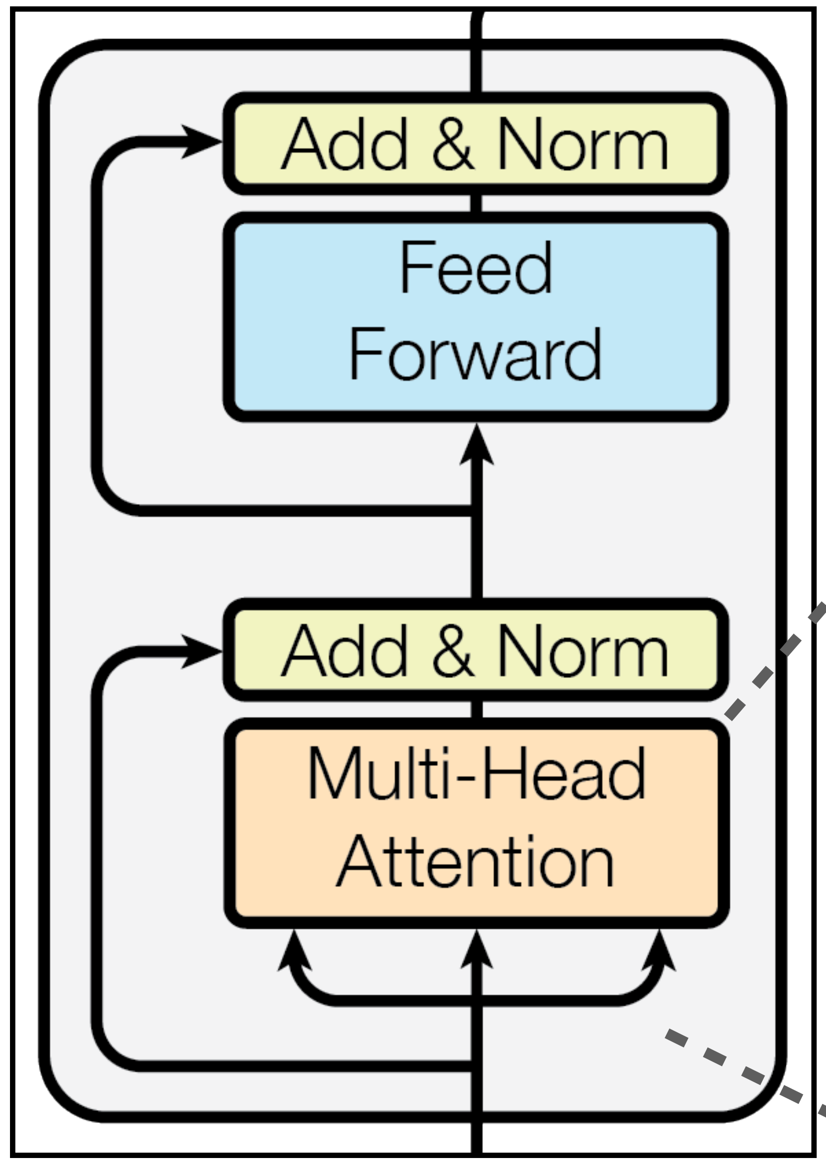
Both input and the output are $\mathbb{R}^D$

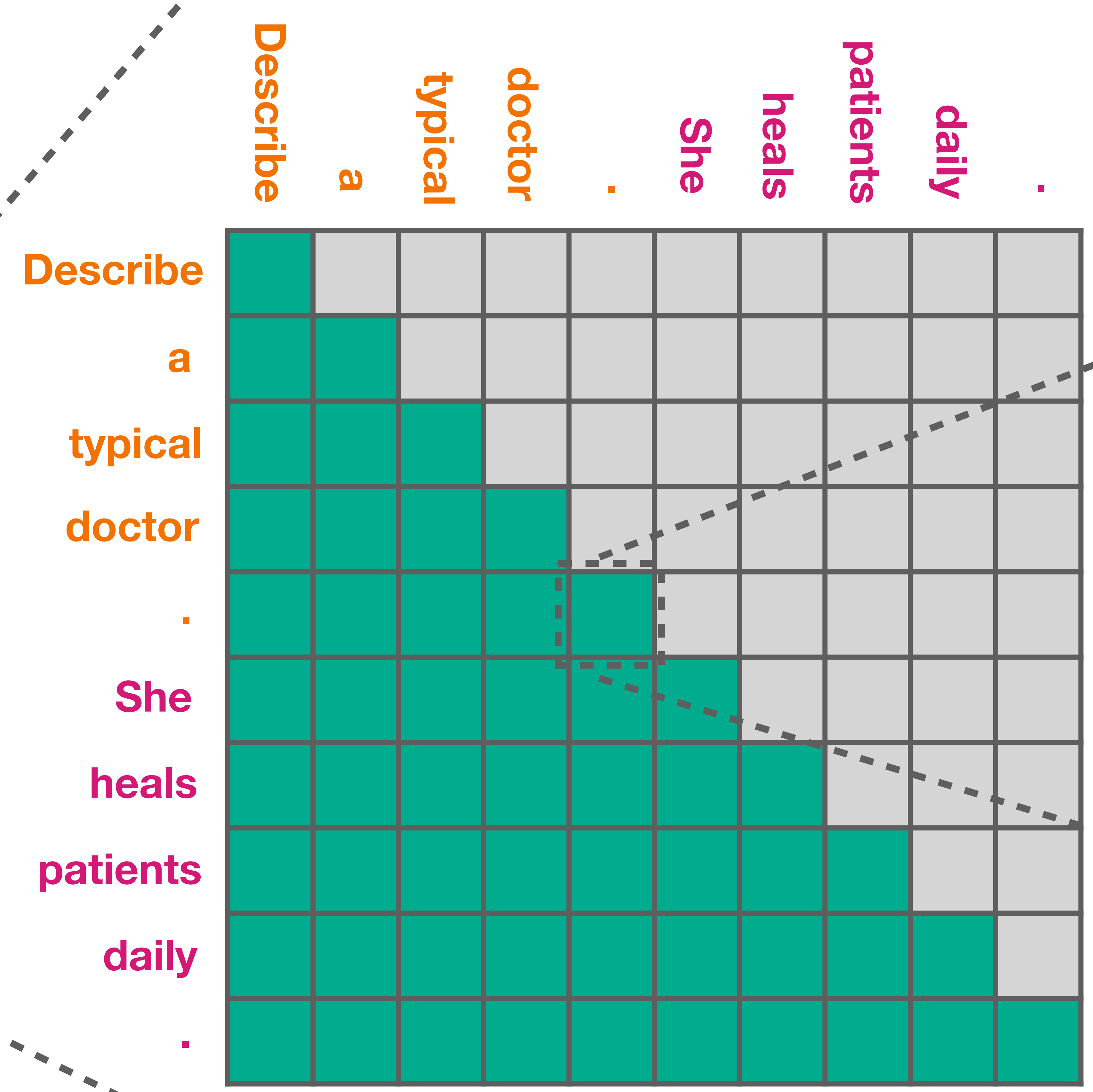
Which part is computationally most expensive?

[Vaswani et al.]

Self attention between all token pairs

When processing *token i*, how much each *token j* should contribute?





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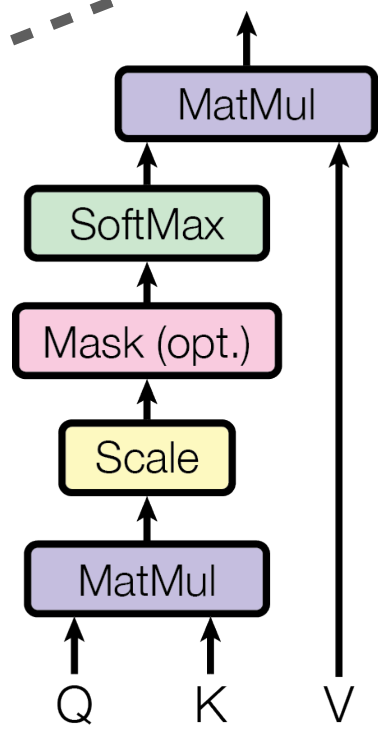
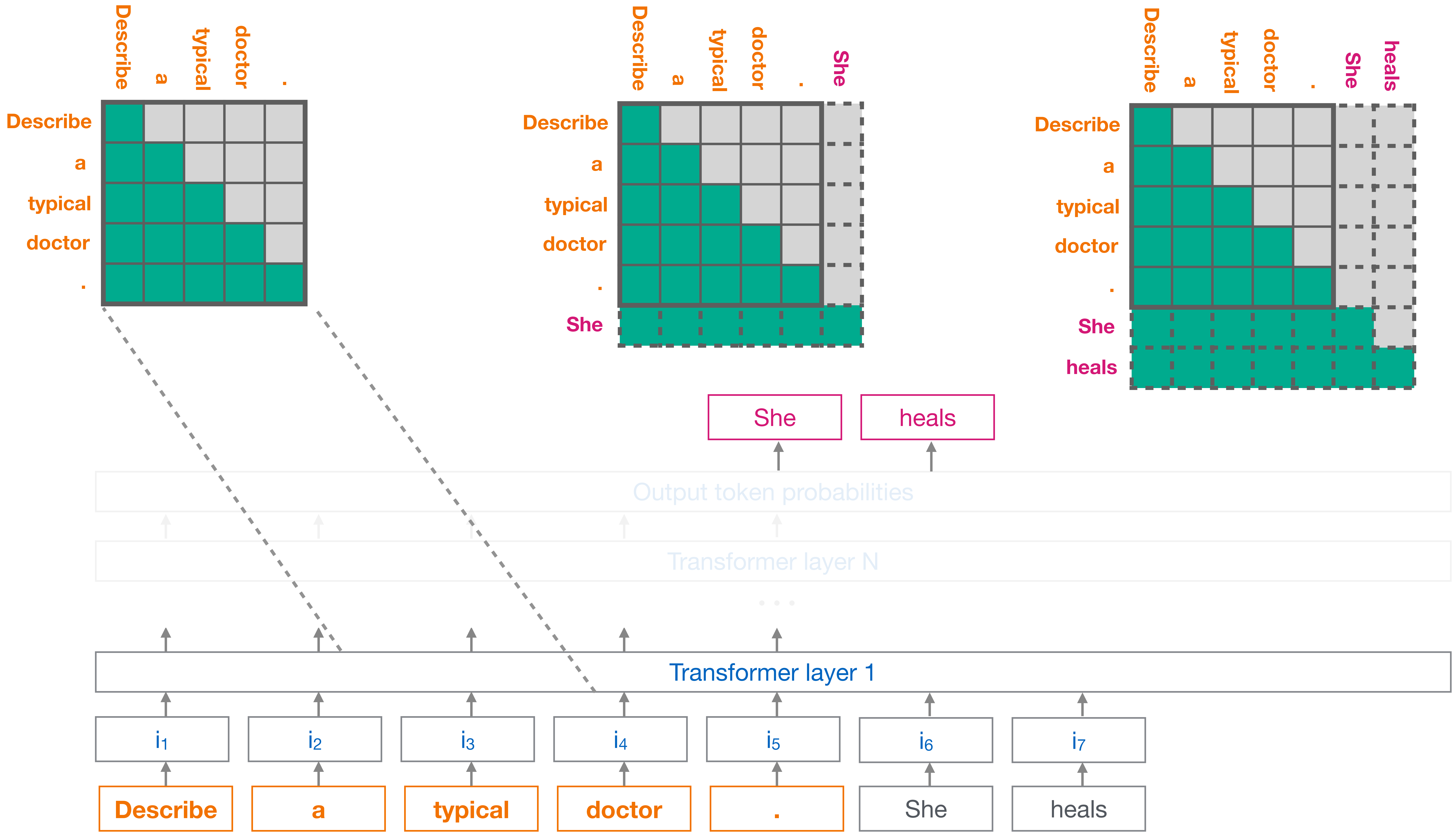


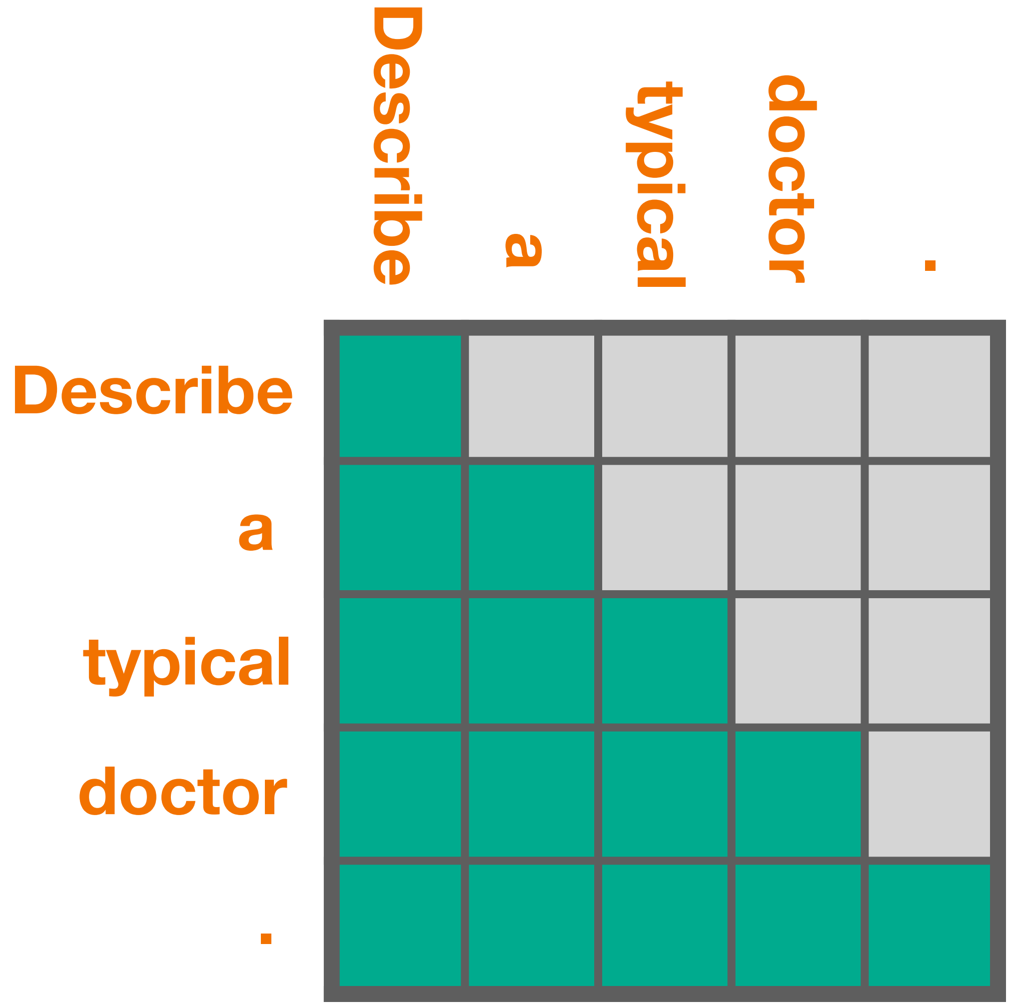
Quadratic in # tokens

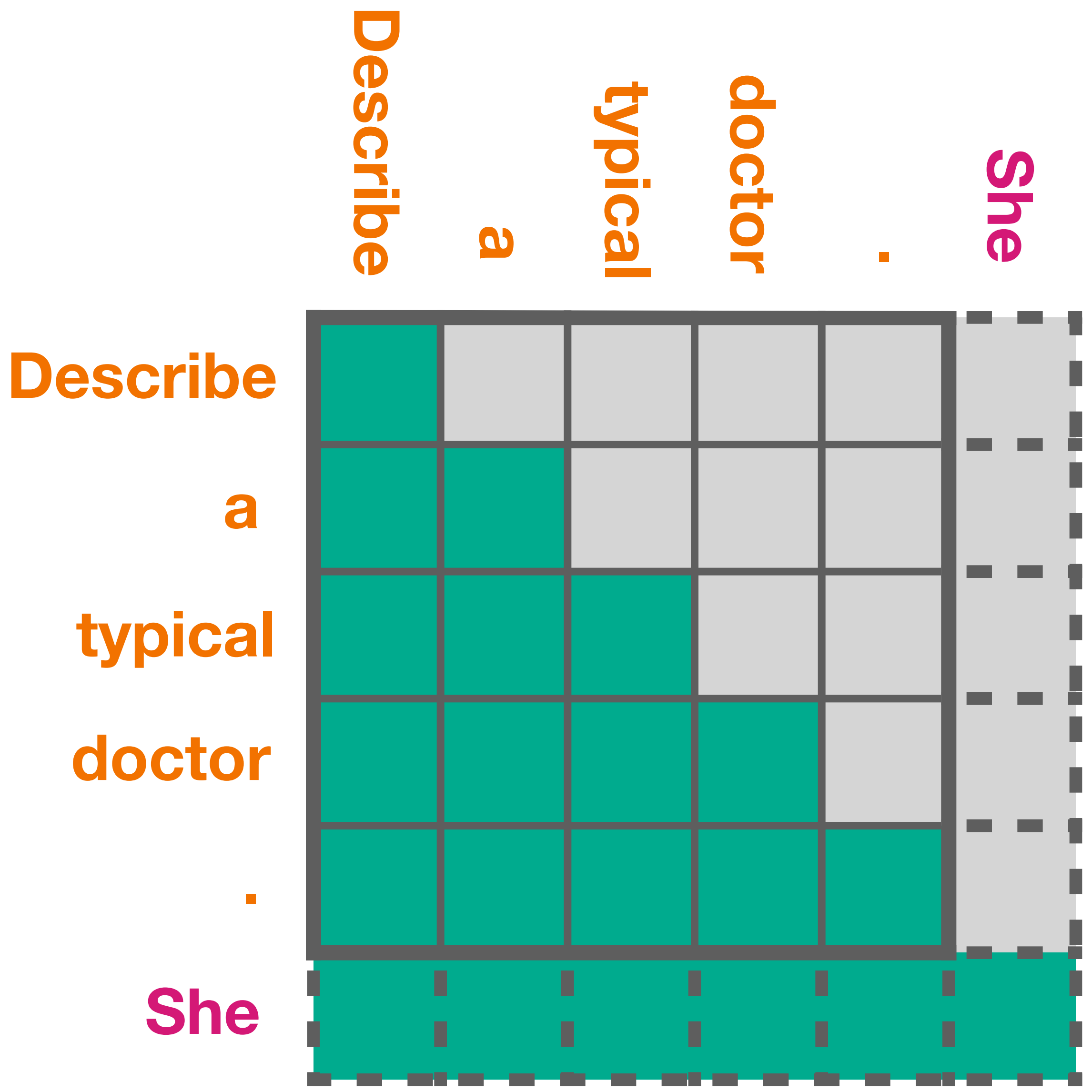
$$\text{attn}(i, j) = \text{softmax}\left(\frac{Q_i K_j^T}{\sqrt{d_k}}\right) V_j$$

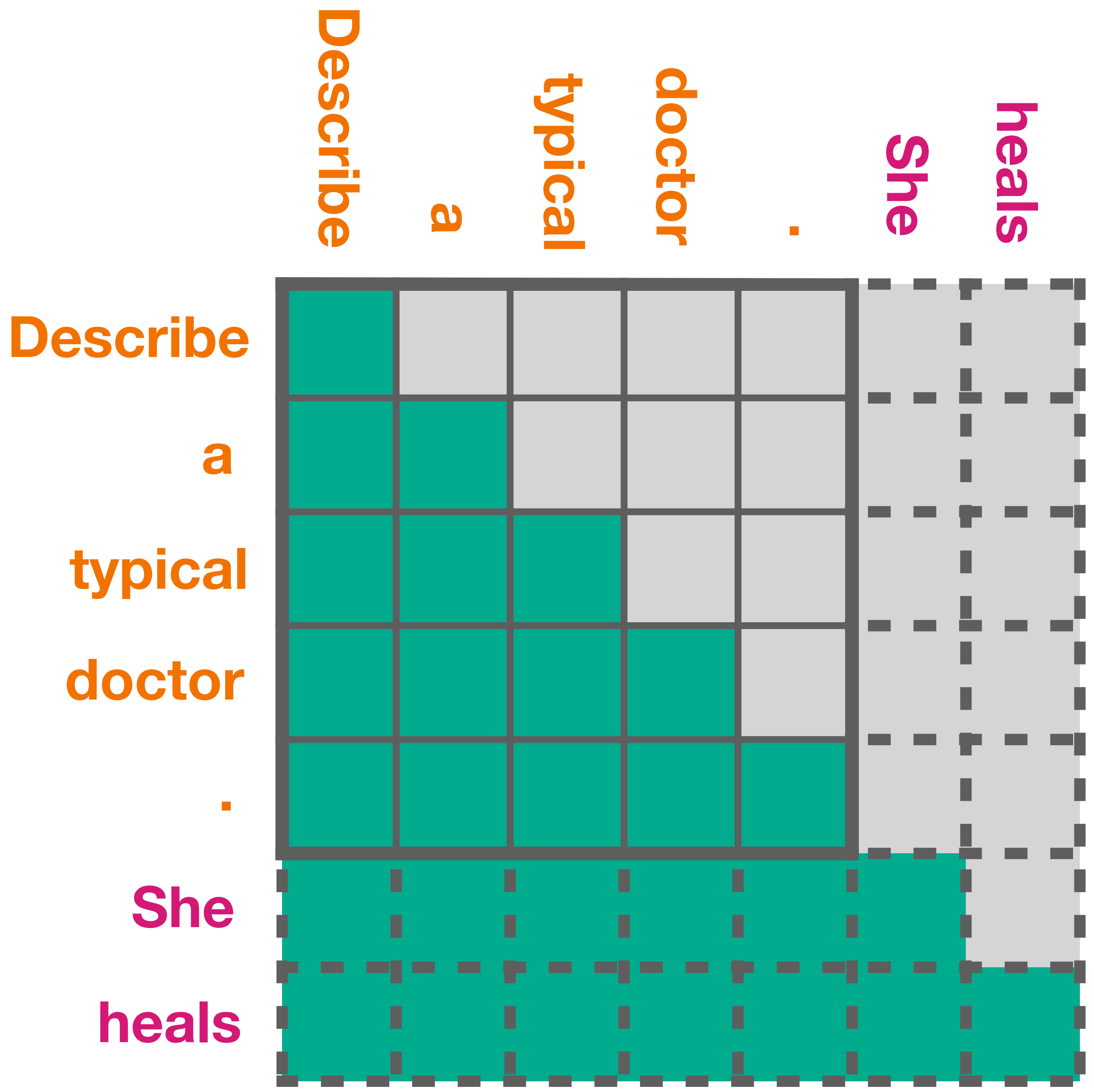
[Vaswani et al.]



Describe
a
typical
doctor
.
She
heals
Output token probabilities
Transformer layer N
Transformer layer 1
i1
i2
i3
i4
i5
i6
i7



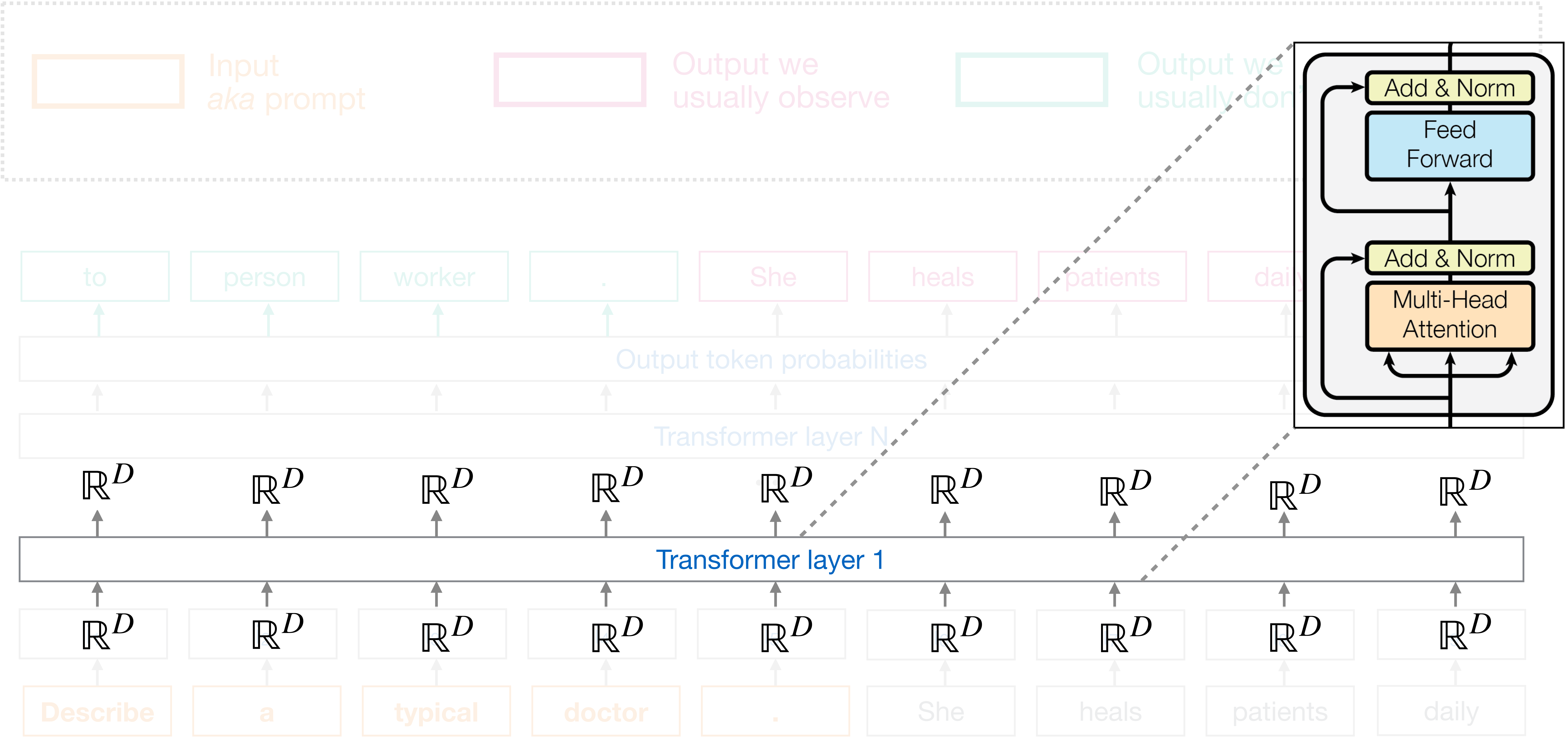




Repeated compute for every generated token

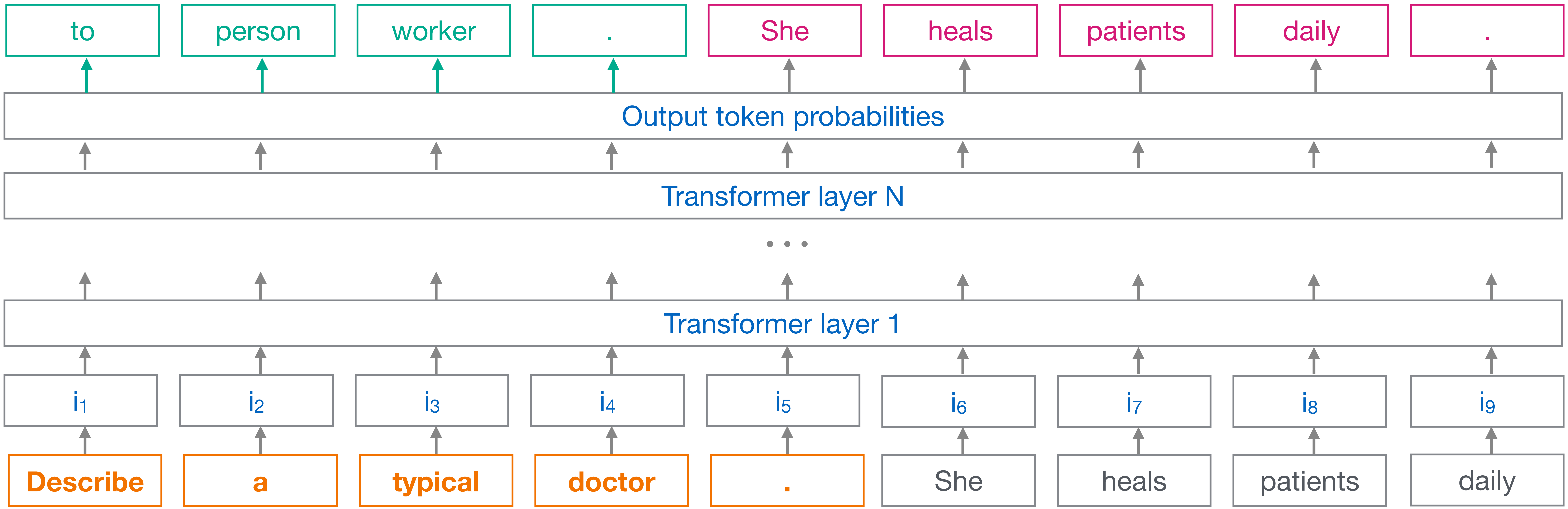
KV-Caching:
Cache the previous attention computation

Recap: What is the output dimensionality of each transformer component?



[Vaswani et al.]

Recap: What is the output dimensionality of each layer?



[Vaswani et al.]

Exercise

Close the Jupyter lab notebook

https://jupyterhub.nrw/

File → Hub Control Panel → Stop my server

References

- Attention is all you need
- Mastering tensor dimensions in transformers
- The illustrated transformer